

Sustainable Spending Rates for Varying Return and Longevity Assumptions

We now have sufficient context to look more closely at the relationship between sustainable spending, investment returns, and time horizon. This analysis will help illustrate the impact of three important matters:

1. What happens to sustainable spending rates as the planning horizon changes,
2. What happens to sustainable spending rates as the fixed investment return changes, and
3. What happens to sustainable spending rates if we build in a constraint to preserve wealth and not accept complete portfolio depletion at the end of the planning horizon.

This analysis will assume inflation-adjusted investment returns, which will allow the sustainable spending rate to be defined as a percentage of the assets available to spend at the initial retirement date, with that spending amount then adjusting for the realized inflation experience in subsequent years. However, readers could treat the investment returns as nominal, in which case the spending rates represent spending that does not adjust for inflation.

Exhibit 2.8 begins the analysis with planning horizons of ten to fifty years, real compounded investment returns of 0 to 6 percent, and an objective of depleting wealth by the end of the planning horizon. In looking at this exhibit, the sustainable spending rates for a 0 percent real return should be intuitive. With no returns, spending 10 percent of initial assets will deplete the portfolio in ten years, spending 5 percent will deplete the portfolio in twenty years, and so on. Without real investment growth, longer time horizons continue pushing the sustainable spending rate lower. As we move down the rows, we see that obtaining returns on the underlying portfolio naturally allows the spending rate to increase. For longer time horizons, spending rates continue to decline, but at a slower rate as returns increase. In other words, long horizons allow for relatively more investment growth to